
CNN-RNN Architectures for sEMG-Based Keystroke Decoding: A Comparison of LSTM and GRU Variants

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Abstract

This paper investigates the classification of the emg2qwerty dataset by comparing a range of LSTM and GRU-based recurrent architectures, including both unidirectional and bidirectional variants as well as hybrid combinations, each built upon a shared CNN frontend. We also evaluated a Transformer encoder (same front-end, then multi-head self-attention and feed-forward layers with CTC) to compare attention-based sequence modeling with recurrent approaches. The dataset consists of preprocessed electromyography (sEMG) signals recorded from wrist activity during typing on a QWERTY keyboard. A hybrid architecture combining Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) was used to decode the temporal sEMG signals into sequences of typed characters. CNN layers extracted local temporal features from the multi-channel EMG signals, while recurrent layers modeled the sequential structure in the signal. The model was trained using the Connectionist Temporal Classification (CTC) loss function, which enables sequence prediction without a direct matching between the number of input signals and output characters. Performance was evaluated using Character Error Rate (CER). This study analyzes how different recurrent architectures influence the model’s ability to capture sequential patterns in the EMG signals and minimize CER while maintaining computational efficiency. Further analysis was done regarding the effects of (1) data preprocessing and augmentation, and (2) variations of the amount of data used within pretraining.

1 Introduction

Capturing temporal patterns is a key challenge in sequential modeling. Convolutional neural networks (CNNs) are effective at extracting local features, they are limited in their ability to model patterns that occur over longer periods of time across sequential data. Recurrent neural networks (RNNs) address this issue by maintaining sequential information across time steps. However, different RNNs vary in how efficiently they capture these patterns and how well they perform in practice.

Our goal is to understand how different sequence modeling architectures affect prediction performance, measured via Character Error Rate (CER). We compare Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, including unidirectional and bidirectional variants, and examine how their update mechanisms influence accuracy and efficiency. We also evaluate a Transformer encoder to assess whether attention-based models can outperform RNNs on sEMG decoding.

By evaluating these architectures with a consistent experimental environment, our goal is to better understand how different sequence modeling approaches influence prediction accuracy and to identify architectural choices that lead to improved performance.

Beyond comparing sequence-modeling architectures, we also studied two practical factors that can substantially affect personalized sEMG decoding performance. First, we evaluated how the amount of user-specific training data influences the Character Error Rate (CER), which helps to determine whether performance differences are driven primarily by architectural choices or by access to additional supervision. Second, we examined several preprocessing and data transformation strategies, including normalization, clipping, a decrease in channels, to measure how input representation and signal conditioning affect decoding accuracy. Together, these experiments complement the architecture comparison by clarifying the data and preprocessing conditions under which these models succeed or fail.

2 Methods

2.1 Architectures

We focused on convolutional–recurrent architectures trained with Connectionist Temporal Classification (CTC) loss, which are well-suited for sequence transcription tasks where the alignment between input signals and output tokens is unknown. Our goal was to explore several CNN–RNN model variants, identify the most effective architecture for sEMG-to-text decoding, and subsequently perform targeted hyperparameter tuning.

We began with the baseline TDS+Conv+CTC model provided in the `emg2qwerty` repository, and ran it for 40 epochs. To better isolate the contribution of the recurrent component, we implemented a simplified architecture consisting of a convolutional front end followed by a recurrent sequence model. The convolutional module contained four convolutional layers designed to extract local temporal and spatial features from the sEMG signals. Because our primary interest was in the behavior of the recurrent module, the convolutional portion was kept fixed across experiments.

We then experimented with different recurrent architectures. The initial implementation used a standard LSTM layer. To allow for fast experimentation, we refactored the model to make the recurrent component modular by introducing an `rnn_type` parameter, allowing the architecture to switch between different recurrent units. Replacing the LSTM with a GRU yielded improved performance, motivating further exploration of recurrent variants.

Next, we introduced bidirectional recurrent layers. Bidirectional models process sequences in both forward and backward directions, allowing the network to incorporate both past and future context when predicting each character (1). This modification is particularly useful for sequence transcription tasks, where the interpretation of a signal segment may depend on surrounding temporal information. We therefore evaluated both bidirectional GRU (BiGRU) and bidirectional LSTM (BiLSTM) architectures.

The same front-end is followed for the transformer architecture by a linear projection to a model dimension d , sinusoidal positional encoding (computed for the actual sequence length to support long test sessions), and a standard Transformer encoder (multi-head self-attention and feed-forward layers). The encoder output is projected to the character set and log-softmax for CTC.

After identifying the best-performing architecture, we performed hyperparameter tuning to further improve performance. The main hyperparameters explored included dropout rate, number of recurrent layers, hidden layer dimensionality, and training duration. The final configuration used a dropout rate of 0.2, two recurrent layers, a hidden size of 512, and training for 60 epochs during model selection on the training and validation sets.

All architectures were trained using CTC loss and evaluated using greedy decoding to produce character sequences from the predicted probability distributions.

After finalizing our architecture, we trained it for 150 epochs for our final results.

2.2 Preprocessing and Transformation Study

The transformation experiments held the model architecture fixed and varied only the preprocessing pipeline. The baseline normalization variant applied per-channel z-score normalization over time before the log-spectrogram transform. Additional variants tested clipping the raw EMG amplitudes to the range $[-5, 5]$ before normalization, adding lightweight or strong augmentation during training, and replacing standardization with robust median/MAD normalization. In the augmentation settings, the repository combines band rotation, temporal alignment jitter, and SpecAugment, with the strong configuration increasing the magnitude and number of corruptions relative to the light setting.

2.3 Channel Count Experiments

We investigated how the number of electrode channels affects decoding performance. Understanding this relationship is important because reducing the number of channels can simplify hardware design, lower computational cost, and improve the practicality of wearable sEMG systems while maintaining acceptable transcription accuracy.

To evaluate this, we trained the same CNN-BiLSTM-CTC architecture described in the previous section while varying the number of input channels. However, for computational efficiency, we used the uni-directional model to see general trends across trials. Channels were removed by selecting evenly spaced subsets of the original electrode array to preserve coverage across the wrist. We evaluated four configurations: the full channel set, 12 channels, 8 channels, and 4 channels.

All models were trained for 40 epochs using the same training and validation splits to ensure comparability across configurations. Performance was measured using character error rate (CER) to quantify how channel reduction affects decoding accuracy.

3 Results

3.1 Effect of Changing Architectures

We evaluated several recurrent architectures using the full 16-channel sEMG setup, including uni-directional and bidirectional LSTM and GRU models, as well as a Transformer baseline. Figure 1 shows the validation CER for all architectures over 60 epochs, starting from epoch 5, as the error from epoch 1-5 saturated the graph, until it settled down to 100 CER.

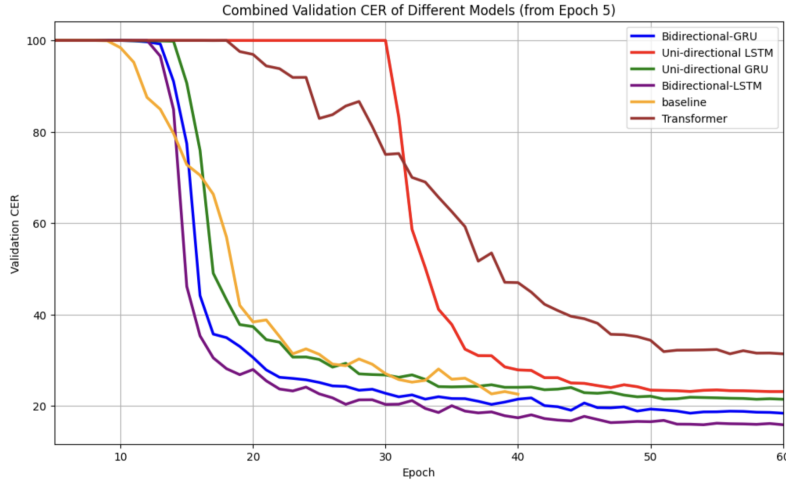


Figure 1: Validation CER for different architectures over 60 epochs (starting from epoch 5, and only 40 epochs for baseline). Bidirectional LSTM achieves the lowest CER.

The results indicate clear differences in both convergence speed and final performance. Table 1 summarizes the final validation and test CERs for each architecture after the initial 60 epochs used for comparison.

Table 1: Validation and test CER for different RNN architectures using the full 16-channel setup. Lower CER is better.

Architecture	Val CER	Test CER
Baseline model (TDS+Conv)	22.52	22.85
Unidirectional LSTM	23.11	23.25
Unidirectional GRU	21.40	21.30
Bidirectional GRU	18.37	18.95
Bidirectional LSTM	15.80	16.96
Transformer	31.47	93.92

To verify full convergence, we trained the BiLSTM model for 150 epochs. Figure 2 shows the training and validation CER curves over the full 150 epochs. The model stabilized after roughly 60–70 epochs, and the final CERs were 15.37 for the validation set and 14.84 for the test set, confirming that the model had fully converged.

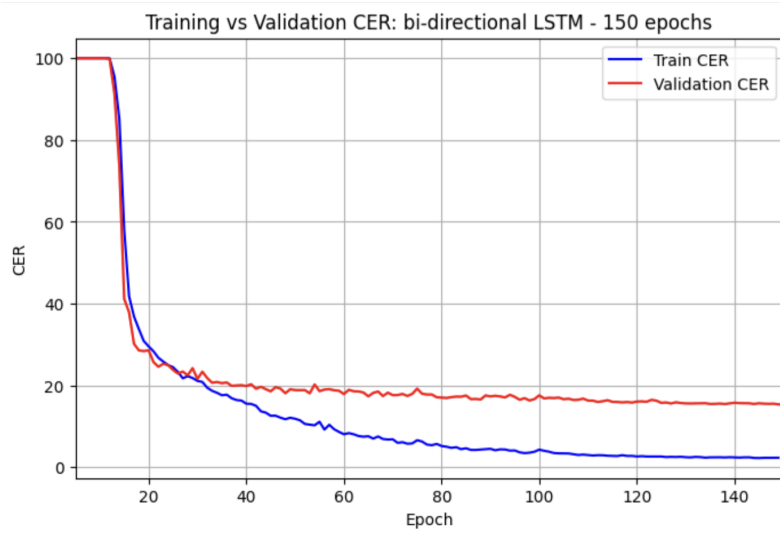


Figure 2: Training and validation CER over 150 epochs for the final bidirectional LSTM model. Validation CER stabilizes around 60–70 epochs; final CER is 15.37 (val) and 14.84 (test) at 150 epochs.

Table 2: Final validation and test CER for the fully trained BiLSTM model (150 epochs).

Model	Val CER	Test CER
Bidirectional LSTM	15.37	14.84

3.2 Effect of Training Data Quantity

Table 3 summarizes the effect of reducing the number of personalized training sessions.

Table 3: Effect of reducing the number of personalized training sessions. Lower CER is better.

Training split	Val CER	Test CER
25% data (4 train sessions)	100.00	100.00
50% data (8 train sessions)	100.00	100.00
75% data (12 train sessions)	19.32	18.89
93.75% data / single_user_90pct (15 train sessions)	19.05	18.46

3.3 Effect of Data Transformations

Table 4 shows the results of preprocessing and augmentation experiments.

Table 4: Effect of preprocessing and augmentation choices. Lower CER is better.

Transformation pipeline	Val CER	Test CER
Per-channel z-score normalization	18.74	19.08
Clamp $[-5, 5]$ + z-score normalization	33.07	29.13
Light augmentation	17.17	16.90
Strong augmentation	33.43	29.54
Median/MAD normalization	17.66	17.94

3.4 Effect of Channel Count

Table 5 summarizes decoding performance with different numbers of sEMG channels.

Table 5: Effect of reducing the number of sEMG channels on decoding performance. Lower CER is better.

Channels	Val CER	Test CER
4 channels	25.27	27.31
8 channels	23.03	23.31
12 channels	23.20	23.20
16 channels	21.95	22.03

4 Discussion

4.1 Recurrent architecture performance

Our results (Table 1) show a clear ranking: bidirectional LSTM achieved the lowest CER (15.37 val, 14.84 test at 150 epochs), followed by bidirectional GRU, unidirectional GRU, and unidirectional LSTM. These differences reveal how temporal context and model capacity affect sEMG decoding. Notably, the unidirectional RNN models performed similarly to the baseline model. Rather, the largest improvements appear when bidirectional context is introduced, suggesting that access to future context is a key factor in improving decoding accuracy.

Bidirectional RNNs leverage full temporal context. By processing sequences forward and backward, bidirectional models exploit both past and future information at each time step. In sEMG typing, key-press patterns span before and after the keystroke itself. For example, the trajectory of the user’s wrist movement after a keystroke can provide cues about which key was pressed. As such, incorporating future context helps reduce errors and accelerate convergence.

LSTM vs. GRU. Unidirectional GRU (21.30 test CER) slightly outperformed LSTM (23.25 test CER), possibly due to simpler gating and reduced overfitting on limited data. In the bidirectional setting, BiLSTM (15.80 test CER) outperformed BiGRU (18.95), showing that LSTM’s extra capacity is better utilized when full sequence context is available.

Model roles. The CNN layers extract local spectral features from the EMG signal, while the RNN layers model how these features evolve over time. Bidirectional recurrence further improves this by allowing the model to use both past and future context when making predictions, which is especially useful for offline decoding. Overall, both the architecture (bidirectional recurrence) and the training strategy (longer training) contribute to improved sEMG decoding accuracy.

4.2 Transformer vs. Recurrent Architectures

We also evaluated a Transformer encoder on the same task: the same spectrogram and multi-band MLP front-end, followed by a linear projection, sinusoidal positional encoding, and a standard

Transformer encoder (multi-head self-attention and feed-forward layers), then a linear projection to the character set with CTC loss. In our experiments, the Transformer achieved validation CER on the order of roughly 30 (depending on hyperparameters), but showed a significant gap between validation and test CER compared to the best CNN+BiLSTM model (15.80 val, 16.96 test). The Transformer tended to overfit more strongly to the training/validation distribution, leading to higher test CER.

Several factors likely explain this. First, Transformers (2) have more parameters and less built-in inductive bias for sequential structure than RNNs; they rely on positional encoding and attention to learn temporal structure from data. With only a single subject’s data, the model may memorize training patterns rather than generalizing. Second, at test time the dataloader feeds full sessions (no windowing), so sequence lengths are much longer than during training. We also used the same preprocessing and augmentation as for the RNN models; Transformers often benefit from different regularization (e.g., higher dropout, weight decay, or more augmentation), which we did not fully sweep. With more data per subject, longer training, or targeted regularization, the Transformer could potentially match or exceed the CNN+BiLSTM, but in our single-subject, limited-data setting, the recurrent architectures proved more robust and achieved the best CER.

4.3 Impact of Training Data Size

Table 3 shows that with 25 or 50 training sessions, the model failed to learn (100 CER on both validation and test). With 75 sessions, performance improved sharply (18.89 test CER), and using nearly the full split (93.75) gave a small additional gain (18.46 test CER). This indicates a *minimum data requirement* for the personalized decoder: below roughly three-quarters of the available sessions, optimization is insufficient and the model does not generalize. Beyond that point, returns diminish. In practice, this suggests that collecting at least a critical mass of typing sessions is necessary before a user-specific decoder becomes useful, and that incremental gains from more data taper off once that threshold is met.

4.4 Impact of Preprocessing and Augmentation

Light augmentation yielded the best test CER (16.90), improving over plain per-channel z-score normalization (19.08). Median/MAD (robust) normalization was also effective (17.94 test CER). In contrast, clamp-based preprocessing (clipping raw EMG to $[-5, 5]$ before z-score) and strong augmentation consistently hurt performance (29.13 and 29.54 test CER respectively). Two takeaways: (1) *moderate* regularization (light augmentation, robust scaling) helps by reducing overfitting and improving generalization; (2) *overly aggressive* perturbations (clamping, strong SpecAugment) distort the signal enough to outweigh any regularization benefit and degrade decoding. The decoder is sensitive to the quality and scale of the input representation; preserving the structure of the EMG signal while adding mild variability is more beneficial than heavy-handed preprocessing.

4.5 Impact of Channel Count

Reducing the number of electrode channels from 16 to 12, 8, or 4 increased CER (Table 5). Using all 16 channels gave the lowest CER; 8- and 12-channel configurations showed similar performance to each other but worse than 16. This suggests that for hardware or cost constraints, a moderate reduction (e.g., 8–12 channels) may be acceptable with a modest accuracy trade-off, but the full 16 channels provide the most informative signal for this single-subject setup. The relationship also underscores that sEMG decoding benefits from spatial diversity across the wrist; removing channels loses information that the model cannot recover.

4.6 Limitations and Future Work

All experiments were performed on a single subject (ID: 89335547), to avoid overloading our limited compute resources. Conclusions about architecture, data quantity, and preprocessing may not generalize to other users or to cross-user settings. Future work could evaluate the same architectures and preprocessing choices across multiple subjects. Comparing the Transformer more systematically against the best CNN+BiLSTM would also give insight into whether attention-based models offer consistent gains for this task.

References

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